Chapter 6 Temporal-Difference Learning

1. What are the two main approaches combined by temporal-difference (TD) learning?

- A. Monte Carlo and dynamic programming
- B. Supervised learning and clustering
- C. Deep learning and search
- D. Planning and supervised learning

2. What is the main difference between TD and Monte Carlo methods?

- A. TD requires a complete model
- B. TD updates estimates after one time step
- C. Monte Carlo never uses experience
- D. Monte Carlo updates only during an episode

3. Which of the following is the TD(0) update target?

- A. The complete observed return
- B. The maximum future action value
- C. The immediate reward plus the discounted next-state value estimate
- D. The average reward across all episodes

4. What does bootstrapping mean in TD learning?

- A. Learning only from terminal states
- B. Using existing estimates to update other estimates
- C. Storing all transitions permanently
- D. Estimating rewards without observing them

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5. Why do TD methods have an advantage over dynamic programming methods?

- A. They always find the optimal policy immediately
- B. They require complete transition probabilities
- C. They do not require a model of the environment
- D. They never need a step-size parameter

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6. What is the primary advantage of TD over Monte Carlo in continuing tasks?

- A. TD can learn without waiting for an episode to end
- B. TD always has zero prediction error
- C. TD requires all possible returns
- D. TD eliminates exploration

7. Which method is an on-policy TD control algorithm?

- A. Q-learning
- B. Sarsa
- C. Dynamic programming
- D. Constant-α Monte Carlo only

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8. Which method is an off-policy TD control algorithm?

- A. Sarsa
- B. TD(0) prediction
- C. Q-learning
- D. Every-visit Monte Carlo prediction

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9. In Q-learning, what is used to determine the next-state contribution to the update target?

- A. The minimum action value
- B. The average of all past returns
- C. The maximum estimated action value in the next state
- D. The first action taken in the episode

Section B. Short-answer questions

1. Define temporal-difference learning and explain its importance in reinforcement learning.
2. What is the prediction problem in reinforcement learning?
3. Explain the relationship between TD learning, Monte Carlo methods, and dynamic programming.
4. Define bootstrapping and explain how it is used in TD(0).
5. What is the role of the step-size parameter α in TD learning?
6. What is the role of the discount factor γ?
7. Explain the difference between a state-value function $v_\pi(s)$ and an estimated value function $V(s)$.
8. What is a sample backup, and how does it differ from a full backup?
9. Why can TD methods learn online and incrementally?
10. What is generalized policy iteration (GPI), and how does it relate to TD control?
11. Explain the meaning of on-policy and off-policy learning.
12. What is the certainty-equivalence estimate, and how is it related to batch TD(0)?

Section C. Comparison questions

13. Compare TD(0) and constant-step-size Monte Carlo learning in terms of their update targets, timing, and bootstrapping.

14. Compare TD learning and dynamic programming regarding their dependence on a model of the environment.

15. Explain the differences between online updating and batch updating.

16. Compare Sarsa and Q-learning in terms of:

- Policy type
- Update target
- Treatment of exploratory actions
- Behavior in the cliff-walking example

17. Compare the predictions produced by batch Monte Carlo and batch TD(0). Why can TD achieve lower prediction error on future data even when Monte Carlo minimizes squared error on the training returns?

18. Compare state-value functions, action-value functions, and afterstate value functions.

Section D. Mathematical and algorithmic questions

19. Write the TD(0) update equation and define every variable.

20. Write the constant-α Monte Carlo update equation. Explain why its target differs from the TD(0) target.

21. Derive the one-step TD target from the Bellman expectation equation for $v_\pi(s)$.

22. Write the Sarsa update equation and explain why the algorithm is named Sarsa.

23. Write the Q-learning update equation. Explain the role of the maximum over next-state actions.

24. Suppose the following values are given:

- $V(S_t) = 4$
- $R_{t+1} = 2$
- $V(S_{t+1}) = 6$
- $\alpha = 0.5$
- $\gamma = 0.9$

Calculate the updated value $V(S_t)$ using TD(0).

25. Suppose $Q(S_t, A_t) = 5$, $R_{t+1} = -1$, $\gamma = 0.9$, $\alpha = 0.2$, and $\max_a Q(S_{t+1}, a) = 7$. Calculate the updated action value using Q-learning.

26. In the random-walk example, the true values of states A through E are $\frac{1}{6}, \frac{2}{6}, \frac{3}{6}, \frac{4}{6}, \frac{5}{6}$. Explain what these values represent and why the center state has value 0.5.

Section E. Application and critical-thinking questions

27. **Driving home example:** Explain how TD learning updates travel-time predictions before reaching home. Why must the Monte Carlo method wait until the journey ends?

28. **Random-walk example:** Describe how TD(0) and constant-α Monte Carlo are evaluated using RMS error. What conclusions does the chapter draw from the results?

29. **Batch learning:** Consider the eight episodes in the predictor example, where state A transitions to B with reward 0 once, and state B has six observed returns of 1 and two of 0. Explain why batch TD(0) estimates $V(A) = V(B) = 0.75$, while batch Monte Carlo estimates $V(A) = 0$ and $V(B) = 0.75$.

30. **Windy Gridworld:** Explain how wind changes the effect of actions in the gridworld. How does Sarsa learn a policy that reaches the goal?

31. **Cliff walking:** Why does Q-learning learn the optimal path along the cliff while Sarsa tends to learn a safer path? Discuss the role of ε-greedy exploration.

32. **Convergence:** Under what conditions does tabular TD(0) converge to the value function of a fixed policy? Distinguish the result for a sufficiently small constant step size from the result under decreasing step sizes.

33. **Afterstates:** Explain why afterstate value functions can make learning more efficient in games such as tic-tac-toe or chess.

34. **Large state spaces:** According to the chapter, why can TD methods be computationally attractive compared with explicitly constructing a maximum-likelihood model and solving for its value function?

35. **Limitations and open questions:** Why does the chapter state that it is not mathematically established that online TD methods always converge faster than Monte Carlo methods?

10. What does Sarsa stand for in terms of the transition sequence?

- A. State, action, reward, state, action
- B. State, average, reward, state, average
- C. Sample, action, return, state, action
- D. State, action, reward, sum, average

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11. Under batch updating, what does batch TD(0) converge to?

- A. The sample-average return estimate in every case
- B. The certainty-equivalence estimate
- C. A random value function
- D. The minimum possible immediate reward

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12. What is an afterstate?

- A. A state before any action is selected
- B. A terminal state only
- C. A state or position after the agent's action has taken effect
- D. A state that has no associated value